DENSO HACKATHON · AI CHALLENGE 2026

Accumulate Master Know-How for AI to Learn

Digitizing expert craftsman know-how on the factory floor — capturing tacit knowledge directly from operational workflow and verbal reasoning without manual documentation.

01. Context → 02. Problem Statement → 03. Case Study → 04. Paradigm Shift → 05. Solution → 06. Architecture → 07. Roadmap

Team: **MentorMind Team**

October 2026

Target: Kitting & Packing Process

— SECTION 01 · FACTORY REALITY

Expert Knowledge Is Silently Leaving the Factory Floor

The looming retirement wave and the deep divide between static documentation and real-world high-precision assembly.

Master Artisans Retiring

Veteran technicians retire, taking with them decades of unwritten, tacit know-how accumulated through hands-on craftsmanship.

SOPs Cannot Keep Up

Bulky manual SOPs are slow to update and fail to capture micro-gestures, subtle hand orientations, and live troubleshooting instincts.

Multimodal VLM + RAG Opportunity

Vision-Language Models and GraphRAG present a historic opportunity to learn and replicate this knowledge — if captured ethically and timely.

DENSO CONTEXT Automotive Precision Manufacturing

High Precision & Craftsmanship Dependence

Denso automotive components require ultra-tight tolerances. Kitting and packing processes heavily depend on technician tactile feedback:

"A slight deviation in placement angle or insertion pressure can trigger cascading defects across an entire production lot."

80%	≤ 5 Mins
Tacit unwritten knowledge	Defect containment window

SECTION 02 · CHALLENGE & DELIVERABLES

Accumulate Master Know-How for AI to Learn

Build an end-to-end system that captures expert workflows and reasoning into a knowledge base, powered by LLMs to reuse tacit knowledge for similar tasks — zero manual documentation required.

DELIVERABLE 01

Standardized Capture Template

Standardized WorkProfile schema (JSON/JSONL) capturing granular steps, key points, underlying reasons ("why"), exceptions, and pass criteria.

`Schema.json` `WorkProfile` `Step-by-step`

DELIVERABLE 02

Hybrid Knowledge Base

Unified graph and vector storage combining Neo4j (equipment-symptom-rootcause-procedure graph) and PostgreSQL pgvector (dense HNSW + lexical RRF).

`Neo4j` `PGVector` `GraphRAG`

DELIVERABLE 03

Dual LLM Assistants

Station-side Work Assistant for apprentice guidance with strict source citations, and RCA Agent for 5-Why root cause investigation linked to Sigma MES.

`Work Assistant` `RCA Agent` `5-Why`

DELIVERABLE 04

Usefulness Evaluation

Quantitative benchmark report powered by Ragas (Faithfulness ≥ 0.85, Relevancy ≥ 0.80) and real-world A/B testing on novice technicians.

`Ragas` `Golden 30+` `A/B Testing`

Suggested Tech Stack

`Multimodal VLM (Qwen3-VL 8B)` `Knowledge Graph (Neo4j)` `PostgreSQL + pgvector` `MediaPipe Hands Pose` `Docling OCR` `FastAPI & Python 3.11`

SECTION 03 · INTERNATIONAL CASE STUDY

When Knowledge Capture Takes an Exploitative Turn

Cautionary lessons from investigative reporting: *"Factory Workers Are Training AI to Replace Them... Here's the Truth"* (Source: youtu.be/5BoTtw8s33g).

COLLECTION METHOD

Passive & Covert Extraction

- Garment and manufacturing workers in India equipped with head-mounted cameras throughout entire shifts.
- Capturing fine-motor gestures and tactile dexterity — real-world "egocentric" data impossible to synthesize in AI labs.
- Targeting low-wage markets (~$240/month) with zero cross-border data protection barriers compared to the US or EU.

CONSEQUENCES & ETHICAL RISKS

Backlash & Erosion of Trust

- Low-wage workers inadvertently train their own automated replacements without consent, transparency, or shared value.
- Fosters workplace paranoia, knowledge hoarding, and active resistance across both blue-collar and engineering ranks.
- Macroeconomic dilemma: If extreme automation displaces human workers entirely, who will have the purchasing power to consume?

> *"Key Takeaway: AI must never be deployed as a covert surveillance tool that destroys worker trust."*

— SECTION 04 · CORE PHILOSOPHY

Two Opposing Approaches: Replacement vs. Empowerment

The fundamental distinction between an AI project that triggers shopfloor resistance and one embraced by factory artisans.

OLD TRAP (CASE STUDY)

Extract & Replace

Exploitation and worker displacement

PURPOSE

Outright worker replacement through aggressive automation

EXPERT ROLE

Passive, covertly monitored, uninformed of data objectives

INCENTIVES

None — basic low wage, zero profit-sharing

ORGANIZATIONAL OUTCOME

Broken trust, knowledge hoarding, ethical backlash, organizational friction

MENTORMIND APPROACH

Capture & Empower

Preserving know-how & empowering artisans

PURPOSE

Preserve, standardize, and amplify master know-how for future generations

EXPERT ROLE

Active co-creator, final approver, attributed and cited in every AI answer

INCENTIVES

Career rewards, reputation benefits, and bonuses for verbal explanations

ORGANIZATIONAL OUTCOME

Knowledge succession secured, collaborative factory culture, proud master artisans

SECTION 05 · COMPREHENSIVE SOLUTION

Non-Invasive Capture + Verbal Explanation Incentives

A streamlined 4-step loop transforming real-world craftsmanship into structured digital intelligence.

1

Non-Invasive Capture

Station camera records 3–5 minute natural work cycles. No change to production flow, zero disruption to cycle times.

2

Explain & Incentivize

The master pauses for 30s to explain "why rotate the part 90 degrees?" → Instantly earns professional recognition and bonus points.

3

Multimodal AI Processing

3-pass VLM pipeline (self-hosted Qwen3-VL) + Docling OCR auto-generates structured WorkProfile drafts. The expert simply reviews and approves.

4

Store & Activate

Ingested into Knowledge Graph (Neo4j) + Vector (pgvector). The expert's name is permanently linked and cited in all AI responses.

Core Differentiator of MentorMind

Experts are **approvers**, not document writers. Experts are **rewarded** for sharing know-how, never covertly exploited.

— SECTION 06 · TECHNICAL ARCHITECTURE

MentorMind 5-Layer System Architecture

Air-gapped self-hosted VLM, Hybrid Retrieval (GraphRAG + pgvector), and Dual AI Agents guaranteeing enterprise-grade data security.

LAYER 1	**Multimodal Ingestion** Station video (4–8 fps), audio commentary, engineering drawings & SOPs	Docling	Station Cam
LAYER 2	**Multimodal VLM Pipeline** 3-Pass loop: Temporal segmentation → Frame description → Task Graph	Qwen3-VL 8B	MediaPipe
LAYER 3	**Hybrid Knowledge Base** Neo4j Equipment-Defect graph + PostgreSQL pgvector RRF	Neo4j	pgvector · Sigma MES
LAYER 4	**Dual AI Agents** Station-side Work Assistant (citations) + RCA Agent (5-Why)	FastAPI	Citations
LAYER 5	**Continuous Evaluation** Ragas quantitative benchmark (Faithfulness ≥ 0.85) + Novice A/B testing	Ragas	A/B KPI

MULTIMODAL INGESTION

Layer 1: Capture & Ingestion

Captures real-world factory data without altering takt time or assembly workflows:

- Station camera: 4–8 fps video sampling (prioritizing high-speed hand gestures).
- Commentary mic: Captures spontaneous artisan verbal explanations.
- Docling / mmore: Structured OCR extraction of engineering drawings and assembly SOPs.

Inspect Full High-Resolution Blueprint

— SECTION 07 · 3-MONTH PLAN & VISION

From Factory Capture to AI Assistant in 3 Months

A pragmatic, milestone-driven execution plan aligned with Denso Hackathon evaluation gates.

MONTH 01

Capture & Standardization

- ✓ Establish standardized WorkProfile template (JSON Schema).
- ✓ Deploy Docling OCR pipeline for drawings & assembly SOPs.
- ✓ Ingest pilot video on Kitting & Packing station (Asian Alliance / Denso).

MONTH 02

Knowledge Base & AI Agents

- ✓ Deploy Neo4j Knowledge Graph + PostgreSQL pgvector (RRF).
- ✓ Ship station-side Work Assistant with strict citation verification.
- ✓ Develop RCA Agent with 5-Why reasoning linked to Mock Sigma MES.

MONTH 03

Evaluation & Pilot Rollout

- ✓ Run Ragas benchmark across Golden Set (30+ validation queries).
- ✓ A/B testing on novice technicians learning station workflows.
- ✓ Latency optimization and packaging final Usefulness Evaluation Report.

"Preserve Master Know-How — Empower Human Artisans, Never Replace Them."

MentorMind stands for human-centric industrial AI: celebrating master craftsmanship, passing on tacit expertise, and empowering the next generation of technicians.

THANK YOU · Q&A